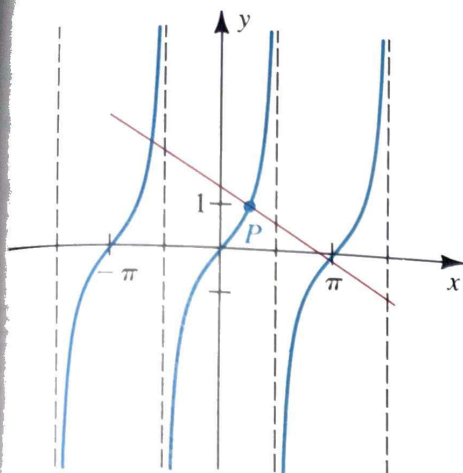


FIGURE 3.24



Using the point-slope form, we can express an equation for the normal line as

$$y - 1 = -\frac{1}{2}\left(x - \frac{\pi}{4}\right),$$

or

$$y = -\frac{1}{2}x + \frac{\pi}{8} + 1.$$

The graph of $y = \tan x$ for $-3\pi/2 < x < 3\pi/2$ and the normal line at P are sketched in Figure 3.24.

When we study *Taylor series* in Chapter 11, it will be necessary to find many higher derivatives of functions. The next example illustrates that these derivatives are easy to find for the sine function.

EXAMPLE 6 Find the first eight derivatives of $f(x) = \sin x$.

SOLUTION Applying (3.25) yields

$$f'(x) = D_x \sin x = \cos x$$

$$f''(x) = D_x \cos x = -\sin x$$

$$f'''(x) = D_x (-\sin x) = -D_x (\sin x) = -\cos x$$

$$f^{(4)}(x) = D_x (-\cos x) = -D_x (\cos x) = -(-\sin x) = \sin x.$$

Since $f^{(4)}(x) = \sin x$, it follows that if we continue differentiating, the same pattern repeats; that is,

$$f^{(5)}(x) = \cos x, \quad f^{(6)}(x) = -\sin x,$$

$$f^{(7)}(x) = -\cos x, \quad f^{(8)}(x) = \sin x.$$

EXERCISES 3.4

Exer. 1–28: Find the derivative.

1 $f(x) = 4 \cos x$

2 $H(z) = 7 \tan z$

3 $G(v) = 5v \csc v$

4 $f(x) = 3x \sin x$

5 $k(t) = t - t^2 \cos t$

6 $p(w) = w^2 + w \sin w$

7 $f(\theta) = \frac{\sin \theta}{\theta}$

8 $g(\alpha) = \frac{1 - \cos \alpha}{\alpha}$

9 $g(t) = t^3 \sin t$

10 $T(r) = r^2 \sec r$

11 $f(x) = 2x \cot x + x^2 \tan x$

12 $f(x) = 3x^2 \sec x - x^3 \tan x$

13 $h(z) = \frac{1 - \cos z}{1 + \cos z}$

14 $R(w) = \frac{\cos w}{1 - \sin w}$

15 $g(x) = \frac{1}{\sin x \tan x}$

17 $g(x) = (x + \csc x) \cot x$

19 $p(x) = \sin x \cot x$

21 $f(x) = \frac{\tan x}{1 + x^2}$

23 $k(v) = \frac{\csc v}{\sec v}$

25 $g(x) = \sin(-x) + \cos(-x)$

26 $s(z) = \tan(-z) + \sec(-z)$

27 $H(\phi) = (\cot \phi + \csc \phi)(\tan \phi - \sin \phi)$

28 $f(x) = \frac{1 + \sec x}{\tan x + \sin x}$

16 $k(x) = \frac{1}{\cos x \cot x}$

18 $K(\theta) = (\sin \theta + \cos \theta)^2$

20 $g(t) = \csc t \sin t$

22 $h(\theta) = \frac{1 + \sec \theta}{1 - \sec \theta}$

24 $q(t) = \sin t \sec t$

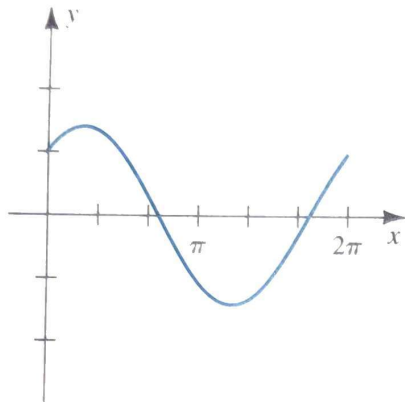
Exer. 29–30: Find equations of the tangent line and the normal line to the graph of f at the point $(\pi/4, f(\pi/4))$.

29 $f(x) = \sec x$

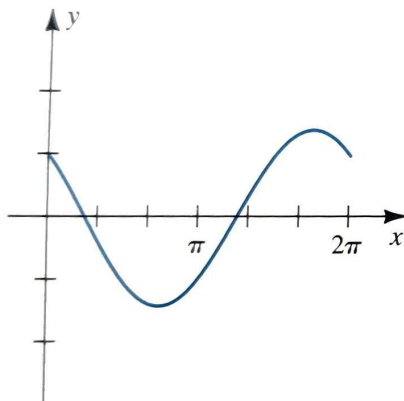
30 $f(x) = \csc x + \cot x$

Exer. 31–34: Shown is a graph of the function f with restricted domain. Find the points at which the tangent line is horizontal.

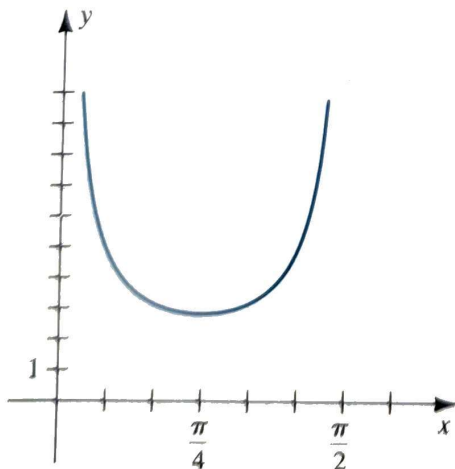
31 $f(x) = \cos x + \sin x; \quad 0 \leq x \leq 2\pi$



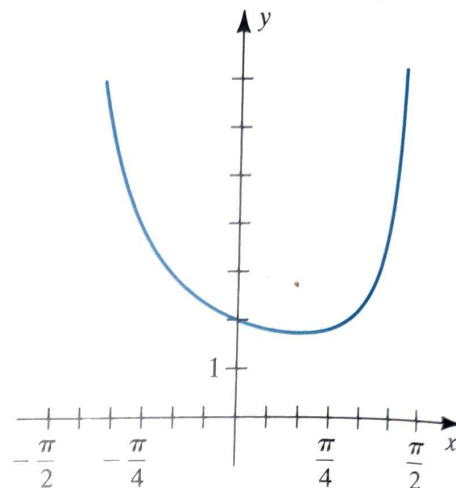
32 $f(x) = \cos x - \sin x; \quad 0 \leq x \leq 2\pi$



33 $f(x) = \csc x + \sec x; \quad 0 < x < \pi/2$



34 $f(x) = 2 \sec x - \tan x; \quad -\pi/2 < x < \pi/2$



Exer. 35–36: **[a]** Find the x -coordinates of all points on the graph of f at which the tangent line is horizontal. **[b]** Find an equation of the tangent line to the graph of f at P .

35 $f(x) = x + 2 \cos x; \quad P(0, f(0))$

36 $f(x) = x + \sin x; \quad P(\pi/2, f(\pi/2))$

37 If $y = 3 + 2 \sin x$, find

[a] the x -coordinates of all points on the graph at which the tangent line is parallel to the line $y = \sqrt{2}x - 5$

[b] an equation of the tangent line to the graph at the point on the graph with x -coordinate $\pi/6$

38 If $y = 1 + 2 \cos x$, find

[a] the x -coordinates of all points on the graph at which the tangent line is perpendicular to the line

$$y = \frac{1}{\sqrt{3}}x + 4$$

[b] an equation of the tangent line to the graph at the point where the graph crosses the y -axis

[c] 39 Graph $f(x) = |\sin^2 x - \cos x \sin(\frac{1}{2}\pi x)|$ on the interval $[0, 5]$ and estimate where f is not differentiable.

[c] 40 Graph $f(x) = \frac{4}{16 \sin 2x - x}$ on the interval $[0, 4]$ and estimate the x -coordinates of points at which the tangent line is horizontal.

Exer. 41–42: A point P moving on a coordinate line has the given position function s . When is its velocity 0?

41 $s(t) = t + 2 \cos t$

42 $s(t) = t - \sqrt{2} \sin t$

Exer. 43–44: A point $P(x, y)$ is moving from left to right along the graph of the equation. Where is the rate of change of y with respect to x equal to the given number a ?

43 $y = x^{3/2} + 2x; \quad a = 8$

44 $y = x^{5/3} - 10x; \quad a = 5$

- 45 [a] Find the first four derivatives of $f(x) = \cos x$.
[b] Find $f^{(99)}(x)$.
- 46 Find $f'''(x)$ if $f(x) = \cot x$.
- 47 Find $D_x^3 y$ if $y = \tan x$.
- 48 Find $\frac{d^3 y}{dx^3}$ if $y = \sec x$.

Exer. 49–52: Prove each formula.

49 $D_x \cot x = -\csc^2 x$

50 $D_x \csc x = -\csc x \cot x$

51 $D_x \sin 2x = 2 \cos 2x$ (Hint: $\sin 2x = 2 \sin x \cos x$.)

52 $D_x \cos 2x = -2 \sin 2x$ (Hint: $\cos 2x = 1 - 2 \sin^2 x$.)

3.5 INCREMENTS AND DIFFERENTIALS

In this section we shall introduce additional notation and terminology that are used in problems involving differentiation. The new notation will allow us to regard dy/dx as a *quotient* instead of merely a symbol for the derivative of y with respect to x . We will also use it to estimate changes in quantities.

Let us consider the equation $y = f(x)$, where f is a function. If the variable x has an initial value x_0 and then is assigned a different value x_1 , the difference $x_1 - x_0$ is called an **increment** of x . In calculus it is traditional to denote an increment of x by the symbol Δx (read *delta x*). Thus,

$$\Delta x = x_1 - x_0.$$

The corresponding increment of $y = f(x)$, denoted by Δy , is

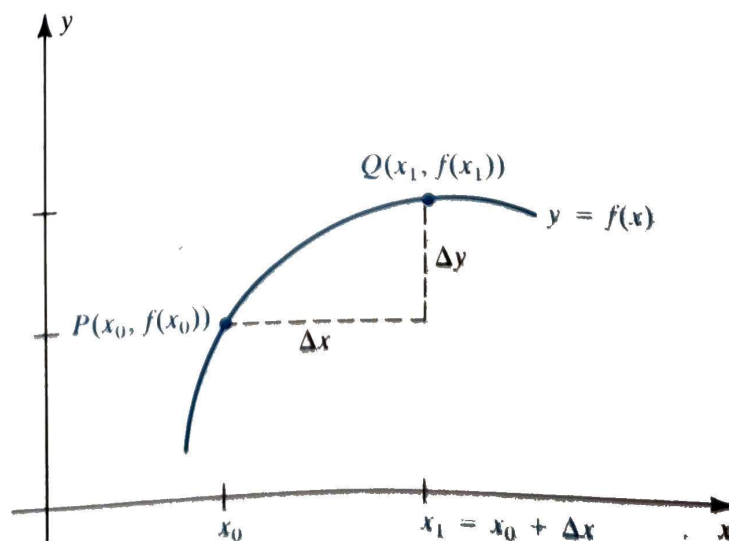
$$\Delta y = f(x_1) - f(x_0).$$

Since $x_1 = x_0 + \Delta x$, we may also write

$$\Delta y = f(x_0 + \Delta x) - f(x_0).$$

The graph in Figure 3.25 illustrates a case in which both increments are positive; however, Δx may be either positive or negative, and Δy may be positive, negative, or zero. In applications Δx and Δy are usually numerically very small.

FIGURE 3.25



If we let x represent the initial value x_0 of the independent variable, then the definition of Δy has the following form.